D07 · Pick-Pack-Ship & Fulfillment Execution

02 · Packing station design & cartonization optimization

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Packing station design organizes materials and flow so orders are packed quickly and correctly; cartonization selects the right box for each order from item dimensions, fragility and carrier dimensional-weight rules.

2 WHY IT MATTERS

Parcel carriers charge on the greater of actual and dimensional weight, so **an oversized box pays freight on air**. An undersized one causes damage and returns. Cartonization fixes both, but only if item dimensions in the master data are accurate — the algorithm cannot compensate for wrong data.

3 HOW IT IS DONE

1. **Clean item dimensions and weights.** Measure the catalog, starting with the top movers; fill missing data.
2. **Rationalize the box set.** Choose sizes from the actual order-cube distribution, and review it annually.
3. **Configure cartonization in the WMS.** Tell the packer which box to use before they reach the station.
4. **Design the station for flow.** Materials within reach, linear or U-shaped layout, scan-to-verify before sealing.

IN PRACTICE

Cartonization only works if all catalog item dimensions are current

The cartonization algorithm selects the box based on catalog item dimensions. If 15% of items have incorrect or missing dimensions, 15% of orders containing those items will be incorrectly cartonized.

Carrier DIM weight cost is the biggest ROI driver for cartonization investment

E-commerce last-mile carriers charge the higher of actual weight and dimensional weight (volume/DIM factor). An oversized box generates a freight cost that can be 2–3× the freight cost for the product's actual weight.

Design the packing station for flow — not packer convenience

A packing station where the operator must turn to take the box, turn again for fill, and turn again for tape is neither ergonomic nor efficient. U-shaped or linear design aligned with the natural process flow reduces packing time and operator fatigue.

Review the box size catalog when the order mix changes

The optimal box size set changes when the order mix changes (NPIs of different sizes, change in average order lines). An annual catalog review keeps cartonization current.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Cartonization implementation — e-commerce company, 3,400 orders/day — Company uses 12 box sizes but packers select the box manually with their own judgment.

Problem identified	Frequency	Impact
Orders packed in box 2 sizes larger than minimum needed (over-boxing)	38% of orders	Unnecessary DIM weight freight cost: +$0.18/order average
Product damage from too-small box (under-boxing)	4.2% of orders	4.2% damage rate on fragile products -> claims and reshipping

Problem identified	Frequency	Impact
Manual box selection time	42 seconds/order	Packing station de-bottlenecking with cartonization implementation

Result: Post-cartonization implementation in WMS: Over-boxing eliminated in 94% of orders. DIM weight freight cost: -$0.18/order × 3,400 orders/day = $612/day = $220K/year. Damage rate: 4.2% -> 1.1%. Box selection time: 42 sec -> 6 sec (WMS indicates the box before the operator reaches the station).

HOW IT IS MEASURED

Cartonization Accuracy % (orders with the correct box per the algorithm)

Cartonization Accuracy %(Orders packed with the box selected by the cartonization algorithm / Total orders packed) × 100

Benchmark: >95% of orders with the correct algorithm-selected box Cartonization accuracy <90% indicates operators are ignoring the system recommendation — possibly because item dimensions in the product catalog are outdated and the algorithm recommends incorrect boxes.

Packaging Cost per Order (packaging materials cost per order)

Packaging Cost per OrderTotal packaging materials cost (boxes, fill, tape, label) / Total orders packed

Benchmark: $0.80–2.50 per order in FMCG e-commerce · Varies significantly by order density and product type Packaging cost per order in an upward trend without a product type or channel mix change indicates cartonization is selecting unnecessarily large boxes or material prices have risen.

WHAT YOU WOULD USE

Cartonization

SAP EWM — Packing Station + Cartonization — Module: WMS-Integrated Cartonization AlgorithmSAP EWM manages automatic box selection with integrated cartonization algorithm. The packing operator receives the box instruction before starting the pack.

Packsize / ProShip — Module: On-Demand Packaging + Cartonization OptimizationOn-demand packaging systems (the box is made exactly the size needed for each order) and world-class cartonization platforms.

4 SO WHAT

Right box, every time, from good data. Fix master dimensions, rationalize box sizes, let the WMS choose and design stations that make the right action the easy one.

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